

$$= 15 \text{ MW}$$

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initial momentum = final momentum

$$mu = (M+m)v$$

gain in GPE = loss in KE

$$(M+m)gh = \frac{1}{2}(M+m)v^2$$

$$h = \frac{v^2}{2g} = \frac{1}{2g} \left(\frac{mu}{M+m} \right)^2$$

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$$\text{efficiency} = \frac{\text{useful output}}{\text{input}} \times 100\%$$